INTEGRATIVE MODELING OF DOXORUBICIN RESPONSE IN CANCER

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BIOE 491: Senior Design Project

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January 13, 2026

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1. SUMMARY

The response of cancer cells to doxorubicin treatment varies significantly due to molecular heterogeneity between cells. These differences are assessed in most studies using static measures summarizing drug sensitivity, but this approach is insufficient to explain the time-dependent dynamics of cell death. This study presents an integrated modeling framework that can address cell-specific differences in doxorubicin response in both dose and time dimensions. First, cell line-specific IC50 values are estimated using gene expression data, and then these static sensitivity measures are used as input to a dynamic model describing time-dependent apoptotic responses. All analyses were performed in the R environment for statistical modeling and ODE-based dynamic simulations. The main objective of this study is to provide a more holistic approach to drug response by examining the relationship between a static metric like IC50 and the temporal dynamics of apoptosis. In this context, the presented approach aims to propose a general modeling framework that brings together static and dynamic components, rather than making a specific numerical performance claim.

2. INTRODUCTION

Doxorubicin is a widely used chemotherapeutic agent that has multiple cytotoxic properties, such as DNA damage, oxidative stress, and mitochondrial toxicity. All these aspects contribute to changes in gene and/or metabolical expression in cells treated with this agent and hence emphasize the biological activity. Doxorubicin is an ideal drug that kills cancer cells due to DNA damage and subsequent p53-mediated apoptosis pathways.Although the agent has displayed potent efficacy in various clinical studies, there are enormous differences in the responsiveness of cells as well as patients to doxorubicin treatment. Doxorubicin is currently a gold-standard agent in various clinical treatments of breast cancers, leukemias, and other solid cancers. The therapeutic index of doxorubicin is really of prime importance because of the severe cytotoxicity associated with its use. For example, its cardiotoxicity has made the dosing of doxorubicin a topic of concern. Cells with identical IC50 values may still follow fundamentally different death trajectories. Drug sensitivity metrics fail to reflect fate heterogeneity.

There is thus an urgent need to comprehend cell- and context-dependent responses in an endeavor to utilize the agent effectively with less toxicity. The cell response to doxorubicin treatment is, therefore, intricately related not only to the dose of the drug administered but also to the cellular properties of molecules and genes. Variations in the level of gene expression among the cells of the cancerous tissues can significantly impact the cell response to the mechanisms of DNA damage and the triggering of the apoptotic pathway.In particular, the mutation status or expression levels of tumor suppressor genes such as p53 are among the critical factors determining the success of the cell death process induced by doxorubicin.

IC50 is one of the indexes that have been developed to quantitatively determine drug sensitivity and is the concentration that results in 50% inhibition of cell viability or proliferation. Due to the advantage of cell comparisons being possible and the relatively low cost involved in acquiring experimentally, the index has found much application in the field of cancer pharmacology. Within the high-throughput screening assay tests, the index acts well as an important indicator that briefly describes the drug sensitivities for various drugs or cell lines. Conversely, the IC50 would be most preferred within drug sensitivity testing linked to the correlation of gene expression profiles/molecular markers.This enables the evaluation of the role behind specific molecular properties with respect to drug sensitivity, addressing, in some

way, cell heterogeneity. From this viewpoint, IC50 emerges as a crucial initial point in the cell-specific drug response summation.

Even if the IC50 value could serve as a very important indicator of drug sensitivity, it remains a static variable. Since the IC50 value relies on cell viability at a given time point, it fails to give insight into the dynamics of drug action on cell viability at various time points. Even for cells undergoing death, the process of death is multi-phasic and occurs at various time points,particularly in DNA-damaging agents such as doxorubicin. In two different cell lines with the same level of IC50 sensitivity, the time of activation of the apoptosis process and rate of death after drug treatment could be different. Some cells would rapidly undergo apoptosis, while others would undergo late or chronic stress adaptation. None of these could be distinguished from the other based on the IC50 value. Therefore, neither static drug sensitivity metrics nor dynamic models alone are sufficient to fully characterize drug response.An integrated framework is required to capture both molecular determinants and temporal response dynamics (Figure 2.1).

The aim of this study is to design an integrative computational framework that links gene expression–based IC50 estimation with time-dependent apoptotic dynamics in doxorubicin-treated cancer cells.

First, the IC50 values are inferred from gene expression data to reflect differences in cell specificity for drug sensitivity. Then, these static values for sensitivity are used as inputs for a dynamic model that reflects the time course of apoptotic responses. This approach enables the use of the differences in the values for the IC50 to model differences in the time courses of cell death responses, for example differences in the rates of apoptosis. This approach seeks to offer a complete representation for doxorubicin-induced cell death by combining the static and dynamic approaches.

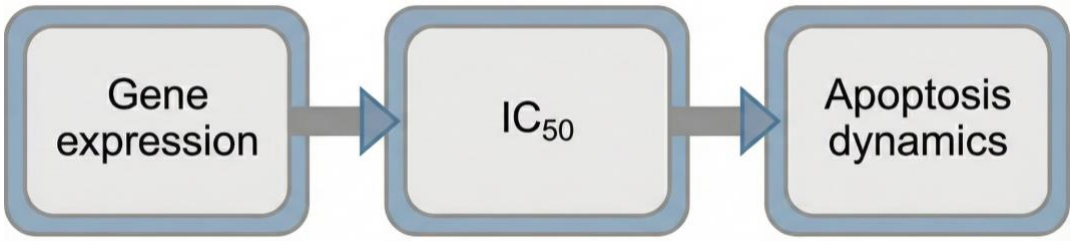

Figure 2.1. Conceptual framework linking gene expression, drug sensitivity (IC50), and apoptosis dynamics.

3. BIOLOGICAL BACKGROUND

Doxorubicin is isolated from the Streptomyces peucetius species and it is an anthracycline antibiotic that is used effectively in different cancer types. From encompassing solid tumors and haematological malignancies to treating both child and adult cancers, it has a broad usage spectrum. Although being highly effective, doxorubicin use is considerably limited due to toxicity because it is nonselective to cancer cells. Additionally, doxorubicin also kills cells through an iron-dependent mechanism of generating reactive oxygen species (ROS), which cause mitochondrial damage and cytochrome c release into the cytosol. Chemically, doxorubicin belongs to the class of anthracyclins. This drug has a tetracycline backbone with a quinone ring and an amino sugar named daunosamine attached through a glycosidic bond. The tetracycline group is water-insoluble, while the daunosamine sugar is responsible for the drug's solubility in water. Doxorubicin is different from daunorubicin, which is another anthracycline that is used commonly, because of the one hydroxyl group that is responsible for the drugs' different spectra of activity. Doxorubicin's antineoplastic activity is caused by a variety of mechanisms, including DNA intercalation and topoisomerase II inhibition, which can cause cell death or growth halt (Meredith & Dass, 2016).

Another cytotoxic mechanism of doxorubicin is the inhibition of topoisomerase II, which targets the topoisomerase II and results in cell death and DNA cleavage. Topoisomerase II is an enzyme that controls DNA's superhelical state, unlinking intertwined DNA strands, as well as relaxing accumulated positive supercoils. Therefore, it is important for completing the process of DNA replication. Topoisomerase II inhibitors, such as doxorubicin, act by stabilizing topoisomerase II-DNA cleavage complexes following topoisomerase II-mediated double-strand breaks in DNA during the replicative process. DNA is then prevented from further processing in the replicative process, resulting in cell death. The residence time of the drug within the complex is an important part of the cytotoxic process for doxorubicin; this is because the topoisomerase II-DNA-drug complex is undone when the drug leaves the complex. Given the fact that the levels of topoisomerase II are higher in proliferating cells, this mechanism of action could make doxorubicin selectively toxic to cancer cells as well. Drug efficacy is dependent on the concentration of the enzyme in the cell because topoisomerase II inhibition requires enzyme-DNA binding to produce cytotoxic action. Studies showing that reduced levels or changed functions of topoisomerase enzymes have produced anthracycline-resistant cancer cells provide additional proof that this is a significant action contributing to doxorubicin's activity.In cells with decreased or modified topoisomerase II, there is also a

corresponding decrease in DNA strand breaks.This implies that topoisomerase II is a crucial component of the medication's effectiveness and that changes in the expression of the topoisomerase gene in cancer cells may lead to drug resistance (Meredith & Dass, 2016).

The p53-status of the cells used for the subsequent experiments was confirmed by the detection of p53 transactivation activity following UV-C and doxorubicin treatment in p53 wt cells but not in p53−/− cells. In addition, an elevation of p53 protein expression was found in UV-C and doxorubicin-treated p53 wt cells. This can be viewed as a common property of p53 induction upon DNA damage. No differences were found between p53 wt cells and p53−/− cells with respect to resistance to doxorubicin. Due to the differences in apoptotic signaling pathways, p53−/− cells would be expected to be less sensitive to chemotherapeutic drugs because of reduced levels of apoptosis induction. p53−/− cells were definitely more resistant than p53 wt cells, though a significant induction of apoptosis was seen in p53 wt cells, which was dose-dependent. Doxorubicin also induces necrosis that is not significant. Thus, the induction of apoptosis, which is enhanced in cells expressing p53, seems to be the cause of the difference in sensitivity between p53 proficient and deficient cells (Dunkern et al., 2002).

Formerly, it was demonstrated that the p53 status affects the cytotoxicity of various anticancer drugs, including anthracyclines, both in vivo and in vitro. However, doxorubicin data are controversial. Therefore, it has been shown that the p53 inactivation causes both sensitivity and resistance of cells to doxorubicin. Because tumors are often mutated in p53, it would be crucial to understand p53's function in doxorubicin-induced cell death in more detail. It was proven that p53 deficient cells are more resistant to doxorubicin than p53 wt cells. This shows that in doxorubicin-induced cell killing, p53 plays a role. Doxorubicin-induced cell death was caused by the induction of apoptosis, which increased with dose. There was barely a slight induction of necrosis. It is clear that p53 influences how doxorubicin affects apoptosis. This is consistent with recent results showing doxorubicin resistance in p53 deletion or mutant cells (Dunkern et al., 2002).

An apoptosis central event is the mitochondria releasing the cytochrome c. Cytochrome c binds in the cytosol with the cytosolic Apaf-1 (apoptotic protease activating factor 1) and procaspase-9 after various pro-apoptotic stimuli. Caspase-9 and main executioner caspase-3 activation will happen and thus, “point of no return” will be reached.The amount of apoptosis-inducing and apoptosis-inhibiting Bcl-2 proteins in the mitochondrial outer membrane regulates cytochrome c release. Inappropriate overexpression of the apoptosis-inhibiting Bcl-2

protein has been revealed to give rise to a wide number of tumor types, including breast cancer. The proapoptotic protein Bax can suppress the resistance to apoptosis conferred by Bcl-2. Tumour cells can develop resistance to all drugs, including all chemotherapeutic agents, irradiation, hypoxic conditions, and growth factor deprivation, if there is an imbalance in the relative expression levels of Bax and Bcl-2 in favour of survival. The upstream regulator of the bax gene is known to be a p53 gene product. This tumor suppressor directly transactivates the transcription of the proapoptotic gene by attaching itself to the bax gene promoter. There is a connection between p53 expression, its mutational status, and changes in Bax expression because wild-type p53 can activate the bax promoter but not mutant p53. When heterodimerized with Bcl-2 and some other members of the Bcl-2 protein family, including Bcl-xL and Mcl-1, Bax's actions seem to be neutralized. The Bax/Bcl-2 ratio appears to be more significant in determining the sensitivity to apoptotic stimuli than the expression of each protein separately, as demonstrated in other tumors (Raisova et al., 2001). The p53-mediated regulation of mitochondrial apoptosis is schematically illustrated in Figure 3.1.

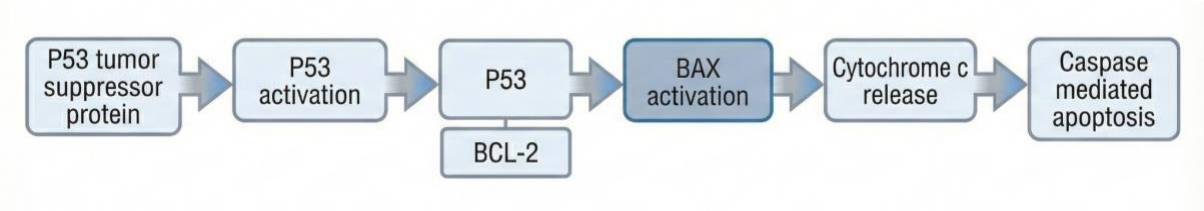

Figure 3.1. Simplified p53-mediated mitochondrial apoptosis pathway

The p53-BAX-BCL-2 pathway was chosen for this study because it is one of the most well-characterized and defining modules in regard to the regulation of apoptosis. Numerous papers have revealed that agents causing DNA damage, for example, doxorubicin, regulate cell fate by targeting this pathway. Though apoptosis can be triggered by several routes in the signaling pathway, it all finally comes to the mitochondrial stage, where the relative levels of pro-apoptotic and anti-apoptic members, namely BAX and BCL-2, respectively, regulate outer mitochondrial permeability. This represents the irreversible step in apoptosis introduced by cytochrome-c release and subsequent activation of the caspases. The p53-BAX-BCL-2 pathway, therefore, represents an essential step for the prediction of apoptosis, as it allows for an easily comprehended explanation for the cellular variation in treatment responsiveness.

IC50 (half maximal inhibitory concentration) is traditionally utilized to define drug potency in cell-based cytotoxicity assays. IC50 is typically utilized as an index aiming to define

the concentration that inhibits 50% cell viability within a definite time interval. Generally, lower IC50 values indicate higher sensitivity to chemotherapeutic drugs. IC50 values tend to differ significantly in different cell lines; this indicates variability in cellular sensitivity to chemotherapeutic drugs. IC50 values were observed to be dependent on cell growth and treatment time. IC50 values were observed to be lower in longer treatment periods. IC50 is thus a static parameter and does not define dynamics associated with cell death. Apoptosis is a time-, phase-, and commitment-dependent biological process. IC50 values thus inadequately define biological times associated with commitment in apoptosis. For this reason, cells demonstrating equal IC50 values may follow different biological dynamics associated with apoptosis. Indeed, IC50 inadequately defines dynamics associated with modulation of cell death pathways. Generally, IC50 is a convenient parameter that inadequately defines dynamics associated with modulation of cell death pathways in cells treated with chemotherapeutic agents (Larsson et al., 2020).

4. METHODOLOGY

This study aimed to study the sensitivity differences of cells to doxorubicin by taking into consideration gene expression profiles and time dependent apoptotic responses together. The study consists of two main parts. In the first part, IC50 values of doxorubicin were estimated using gene expression data. In the second part, the possible effects of different IC50 levels were evaluated comparatively on the time dependent apoptotic response. This approach is consistent with the general framework used in gene expression-based drug sensitivity prediction studies (Community et al., 2014).

4.1. Overall Workflow

The overall workflow of this study is based on predicting drug sensitivity from gene expression data and evaluating these predictions in relation to time-dependent apoptotic responses. Firstly, a dataset with gene expression data and IC50 values of doxorubicin was used. Based on these data, IC50 values were estimated from gene expression profiles in R. The estimated IC50 values were grouped to represent different scenarios in which cells showed low, intermediate, and high sensitivity to the drugs. These scenarios were used to examine how cellular responses can change over time. Then, IC50 scenarios were used as input to the ODE constructed in R. This model represented the time-dependent course of apoptotic response and how the response changed with different IC50 values. Lastly, the time point at which the apoptotic response reached a specific threshold was defined as T50, and the relationship between drug sensitivity and time-dependent cellular response was evaluated by comparing T50 values obtained for different IC50 scenarios.

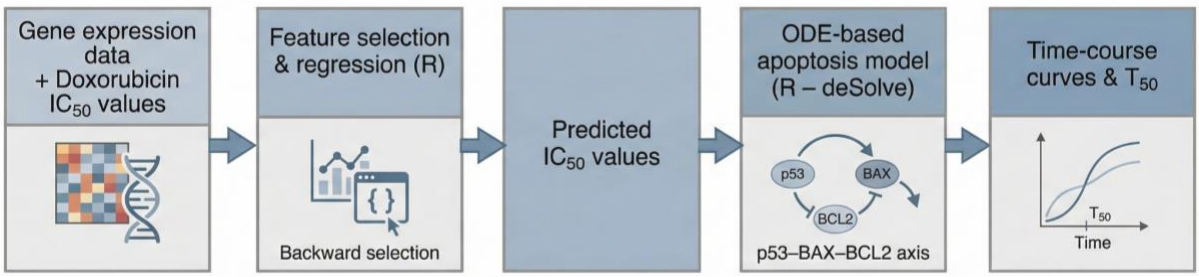

Figure 4.1.1: Simplified overall workflow of computational drug response and apoptosis

4.2. R-based IC50 Prediction from Gene Expression

In this study, gene expression data of cancer cell lines were used with IC50 values to represent the sensitivity of these cells to doxorubicin. Gene expression profiles were used as input variables, while IC50 values were defined as the output variable.

Due to the large size of gene expression data, backward elimination was preferred for modeling because it has mild significance level criterion and it is superior to other approaches for bias. Backward elimination removes variables that contribute limitedly to the model one-by-one (Dunkler et al., 2014).

The performance of the established models was evaluated by using the coefficient of determination (R2) and mean absolute error (MAE) metrics. R2 was used to see how well the model explains the variability of IC50 values, providing a measure of overall predictive performance. MAE provides an assessment for the model's error level by measuring the mean difference between predicted and actual IC50 values (Chicco et al., 2021).

The most suitable model was selected by considering both R2 and MAE values obtained by applying cross-validation because cross-validation can be successfully used for comparing predictive models and selecting optimal model configurations (Bates et al., 2021). Models with higher R^2 and lower MAE values were preferred because they offered a more balanced result between prediction performance and error level. The model that best met these criteria was selected to use in analyses.

4.3. ODE-based Dynamic Apoptosis Modeling in R (deSolve)

Apoptosis is a dynamic process resulting from the time-dependent interaction of intracellular signaling pathways. Activation and suppression levels of signaling proteins change over time determines the progression of apoptotic response (Mustafa et al., 2024). Therefore, differential equations were used to study the time dependent change of the apoptotic response.

In this study, the apoptosis process is represented by a simplified differential equation model constructed using the deSolve package in the R. Instead of including all the details of the biological process, the model has been simplified to include only the essential components known to play a central role in the decision-making process regarding apoptosis. This approach was chosen to improve the model's clarity and to compare different scenarios.

The model is based on the p53-BAX-BCL2 axis. In response to DNA damage, p53 gets activated. Activated p53 directs the apoptotic process through increased pro-apoptotic BAX and suppressed anti-apoptotic BCL2 (Haupt et al., 2003). Therefore, the apoptotic response was evaluated through the time-dependent change in the BAX/BCL2 ratio.

IC50 values obtained from gene expression data in R were used in the model as a parameter representing the cell's sensitivity to drugs. Different scenarios representing low, medium, and high sensitivity were created based on IC50 values, and these scenarios were considered as inputs to the model. Thus, the effects of different IC50 levels on the time-dependent course of the apoptotic response were examined by comparison.

The aim of this modeling approach is not to numerically predict actual biologic times or protein levels accurately. The goal is to demonstrate, within comparative framework, how different levels of drug sensitivity can influence the time-dependent behavior of the apoptotic process.

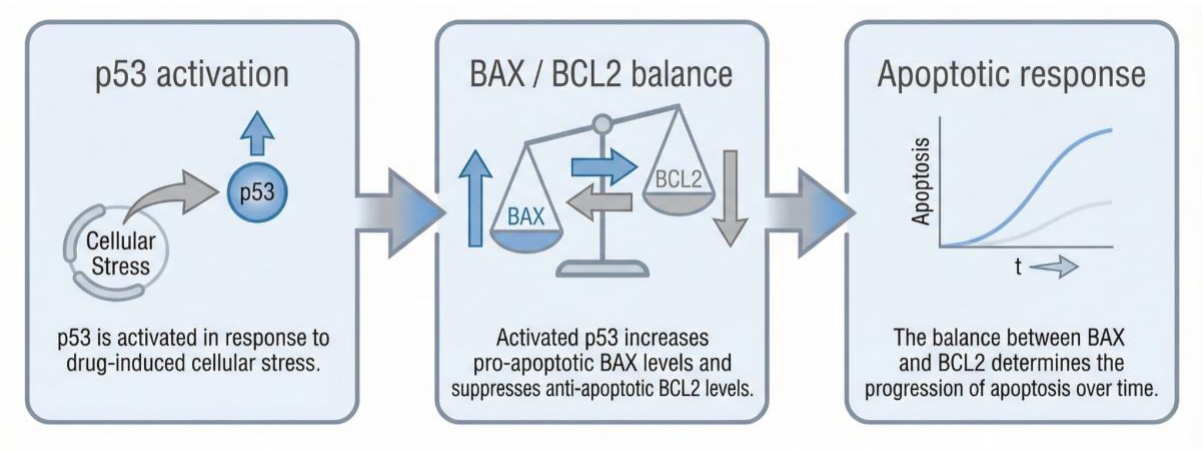

Figure 4.3.1: Simplified dynamic model of p53 mediated apoptotic response

5. CONCLUSION

In this study, an integrated modeling approach which combines static drug sensitivity measures with time-dependent apoptotic processes is developed for a more comprehensive investigation of cellular response to doxorubicin. The main goal of this study is not to create numerically precise predictions, but rather to reveal the conceptual relationship between the gene expression based IC50 predictions and the dynamic nature of apoptosis.

The usage of IC50 values gained from gene expression data as input to a simplified apoptosis model defined by differential equations, shows how static drug sensitivity measures can be related to temporal biologic responses. This integration emphasizes that drug response is not only a dose-dependent fact, but also a dynamic process of intracellular decision mechanism that shapes over time. This approach presents conceptually that even cells with similar IC50 values may exhibit different behaviors in terms of the starting time and progression of apoptosis.

Model built on p53–BAX–BCL2 axis,biologically meaningful decision point for irreversible apoptosis and purpose to represent interpretable frame.In this content IC50, rather than defining alone a cell destiny, takes over as an organizer parameter for apoptotic period dinamics.In this perspective, it puts forth traditional static sensitivity criterion.

Though the model has been kept simple in this context, it is possible in the future to extend the model with increased complexity. Moreover, it can be applied for varied drugs and cancers. Finally, it can be concluded that this study provides a flexible and complete way for studying the effects of molecular heterogeneity in time-dependent responses to therapy and helps in better understanding drug sensitivity in cancer.

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